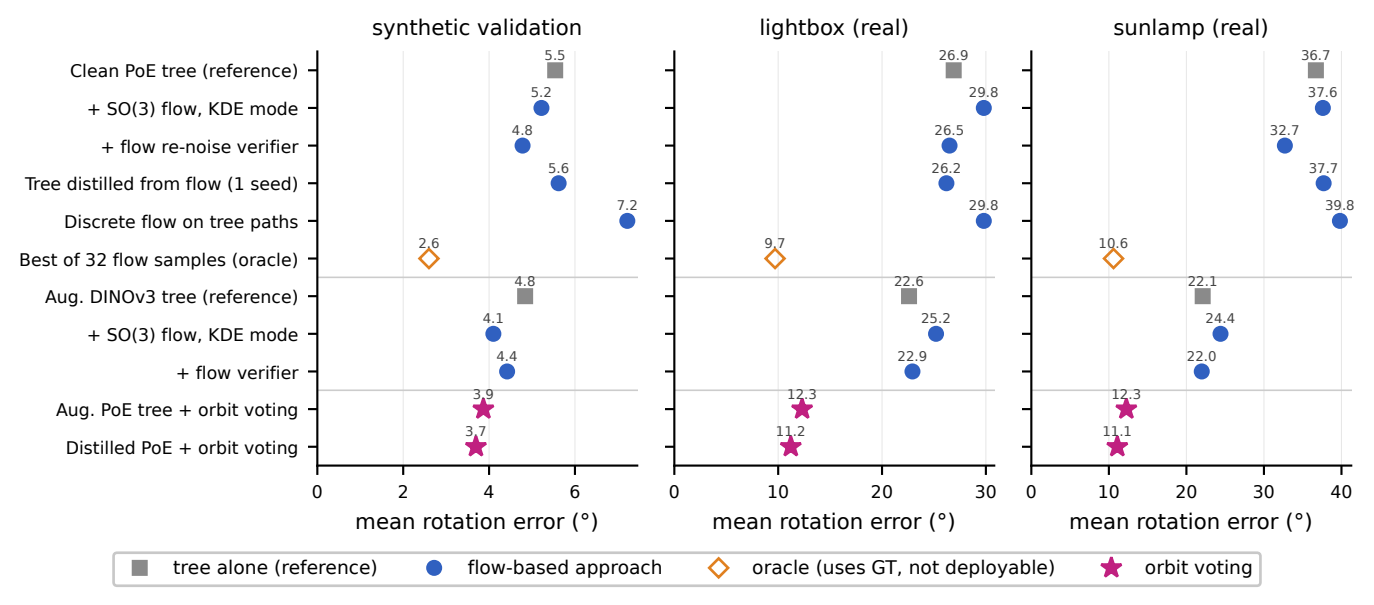
synthetic validation
lightbox (real)
sunlamp (real)
Clean PoE tree (reference)
+ SO(3) flow, KDE mode
+ flow re-noise verifier
Tree distilled from flow (1 seed)
Discrete flow on tree paths
Best of 32 flow samples (oracle)
Aug. DINOv3 tree (reference)
+ SO(3) flow, KDE mode
+ flow verifier
Aug. PoE tree + orbit voting
Distilled PoE + orbit voting
5.5
5.2
4.8
5.6
7.2
2.6
4.8
4.1
4.4
3.9
3.7
26.9
29.8
26.5
26.2
29.8
9.7
22.6
25.2
22.9
12.3
11.2
36.7
37.6
32.7
37.7
39.8
10.6
22.1
24.4
22.0
12.3
11.1
0
2
4
6
0
10
20
30
0
10
20
30
40
mean rotation error (°)
mean rotation error (°)
mean rotation error (°)
tree alone (reference)
flow-based approach
oracle (uses GT, not deployable)
orbit voting